

GENERAL SCIENCE — GRADE 10 | NUTRITION IN ANIMALS

Quick Check

Lesson 7: Model Building — Making a 3D Model of the Human Digestive System

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Nutrition in Animals | Lesson 7

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A learner sketches the journey of a piece of ugali through the alimentary canal. Which sequence of organs is correct?

A Mouth → oesophagus → stomach → large intestine → small intestine → rectum

B Mouth → oesophagus → small intestine → stomach → large intestine → rectum

C Mouth → oesophagus → stomach → small intestine → large intestine → rectum

D Mouth → stomach → oesophagus → small intestine → large intestine → rectum

A group is writing its building list for the mouth. Which entry is correct for the enzyme and substrate in the mouth?

- A** Salivary amylase acts on proteins to form peptides
- B** Lipase acts on fats to form fatty acids and glycerol
- C** Salivary amylase acts on starch (as in ugali) to form maltose
- D** Pepsin acts on proteins to form peptides

A group is building its model and is asked: what does bile do to the fat in nyama choma, and why is that important?

- A** Bile carries digested fat across the villi into the bloodstream
- B** Bile emulsifies fat into tiny droplets, giving lipase a larger surface to act on
- C** Bile is an enzyme that chemically splits fat into fatty acids, so lipase is not needed
- D** Bile is made in the stomach and digests the proteins in the meat

A group labels the large intestine on its model. Which label is correct?

A Water is reabsorbed here; no digestive enzymes are made here

B Pepsin digests proteins into peptides

C Maltase and sucrase finish the digestion of sugars

D Lipase digests the remaining fats

Damp sponge strips are placed inside the tube that represents the small intestine. What do these strips represent, and why do they matter?

- A** Bile, which breaks up fat into small droplets
- B** Sphincters, which stop food from moving backwards
- C** Stomach walls, which churn the food mechanically
- D** Villi, which greatly increase the surface area for absorbing nutrients into the blood

At the start of the unit, a learner said, "The stomach digests everything in our ugali, sukuma wiki and nyama choma." Which statement best shows where this idea is wrong?

- A** The stomach digests starch fully, so the small intestine only absorbs water
- B** The stomach only stores food, and no digestion happens there
- C** The stomach makes bile, so it digests fats but not proteins
- D** The stomach starts protein digestion with pepsin and HCl, but most chemical digestion and nearly all nutrient absorption happen in the small intestine

A group adds a red 'Failure Mode' tag to the pancreas on its model. Which tag correctly describes what happens if the pancreas stops producing lipase?

- A** Starch in the ugali cannot be digested in the mouth
- B** Proteins in the meat cannot be digested in the stomach
- C** The fat in nyama choma is not digested and passes into the large intestine unabsorbed
- D** The large intestine cannot reabsorb water from the waste

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).